

PRINCE MATHUR

SENIOR SOFTWARE ENGINEER

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SUMMARY

Senior Software Engineer with 7.5+ years of experience designing, scaling, and owning distributed systems across Fintech, EdTech, and Restaurant Tech industries. Proven track record leading engineering teams of 5+ engineers for 2.5+ years, driving end-to-end architectural decisions, and delivering production-grade systems at scale. Experienced in async processing, event-driven architecture, microservices, and AI-native integrations.

SKILLS

- **Languages & Frameworks:** Python, Flask, FastAPI, Celery, Ruby on Rails, Sidekiq, JavaScript, React, Node.js
- **Databases & Cloud:** MySQL, PostgreSQL, MongoDB, Redis, Elasticsearch, AWS (S3, EC2, Texttract, Cognito, Elastic Beanstalk, Chalice, Secrets Manager), Azure (Blob Storage, Key Vault), Docker, Git
- **System Design:** REST API Design, WebSockets, Real-Time Systems, Distributed Systems, Event-Driven Architecture, Async Processing, Microservices, PCI Compliance
- **AI & Automation:** LangChain, NLP, OpenAI API, AWS Comprehend, Cursor, Windsurf, Claude Code
- **Delivery & Collaboration:** Unit Testing, Performance Optimisation, Code Reviews, CI/CD (GitHub Actions), Jira, Technical Mentorship, Cross-Functional Collaboration

EXPERIENCE

Senior Software Engineer

Apr 2025 - Apr 2026

Checkmate 🔗 (Restaurant technology company)

New York, USA (Remote)

- >\$100K in inflated point balances across **1,744 users** and 4 brands by re-engineering the core credit-debit transaction matching logic and building an **idempotent batch-processing pipeline** to restore historical integrity with zero downtime.
- Contributed to **FreedomPay** payment integration, implementing a **PCI-compliant** system that reduced unknown-state payment failures, earning FreedomPay certification and unlocking 5+ restaurant brands.
- Built a high-throughput marketing events ingestion pipeline processing up to **1,000 notification IDs** per campaign, improving analytics across all brand campaigns.
- Delivered a customer loyalty and marketing data pipeline covering **100K+ customers** across 6 restaurant brands at a **99.3% success rate** over tens of thousands of records.
- Architected an **async, event-driven** order-rating system for the Kiosk backend, decoupling rating submission from order creation to enable targeted product improvements from customer feedback.
- Developed a **dual-mode surcharge calculation** feature for configurable pre/post-discount logic and a store-level admin toggle for accurate tax and fee reporting across brand and revenue-centre levels.
- Established **AI-assisted development workflows (Cursor, Windsurf)** with structured context files for session consistency, accelerating boilerplate, test generation, and PR reviews across the team.

Senior Software Engineer (Tech Lead, Neovarsity)

Mar 2022 - Apr 2025

Scaler Academy 🔗 (Education platform)

Bengaluru, India (Hybrid)

- Served as **Tech Lead for Scaler Neovarsity** for 2.5 years, managing a **team of 5 engineers**, owning system design, architecture decisions, sprint planning, standups, and retrospectives. Acted as primary technical point of contact for the product manager, program manager and engineering manager.
- Led cross-functional collaboration with **Woolf University** (external partner) from integration through platform launch, leading weekly requirement and alignment calls, and coordinating across internal teams to resolve dependency blockers and align on shared API contracts.
- Architected a scalable data sync framework in **Ruby on Rails + Sidekiq** with idempotent job design, processing **40M+ jobs** and enabling real-time sync for **55K+ users**, eliminating manual process overhead of CSV exports.
- Engineered high-throughput job processing at **500+ jobs/min** using Sidekiq queue throttling, circuit breakers, and idempotent batch processing, handling **3M+ activities** within 10 days.

- Designed DB schema and systems for automated grading, credit reconciliation, and curriculum redesign, directly contributing to graduation of **300+ degree students**.
- Eliminated instructor no-shows for **3+ years** by building a **multi-channel alerting system (Kaleyra Voice calls + Slack)** with layered escalation across **50+ daily classes**, guaranteeing **100% class coverage** every day.
- Drove SQL query performance optimisation across the system and built **Metabase dashboards** with automated alerts for real-time system and business monitoring across different features.
- Served as rotating on-call system admin, monitoring **Jenkins / AWS Elastic Beanstalk** deployments, triaging queue flooding and slow query spikes via Sentry / DB CPU metrics, and leveraging AWS Auto Scaling Groups and Load Balancer for traffic reliability.
- Migrated from Mentor-based mock interviews to an AI-driven mock interview microservice for **40K+ learners (₹30L saved to date, ₹1.25Cr projected annually)**.
- Mentored 5 engineers via PR reviews, LLD critiques, and 1:1s; conducted intern hiring interviews; ensured **100% RSpec test coverage** across all backend features.

Full Stack Developer - Level 2

Apr 2019 - Feb 2022

Egregore Labs  (AI-driven fintech solutions)

New Delhi, India (Hybrid)

- Led backend development for Romulus, a trade finance POC that won **Finastra's #HackToTheFuture hackathon** and reached the finals across 3 categories internationally.
- Engineered multi-source document ingestion pipelines using **AWS Textract, Azure Text Extractor, OpenCV, spaCy, and AWS Comprehend**, supporting **50+ document templates** across 10,000+ samples with provider-level abstraction and improved checkbox detection via OpenCV.
- Developed a Python package abstracting cloud service integrations across **AWS, Azure, and GCP**, improving consistency and reusability across all microservices.
- Implemented keyword-based filtering using **Elasticsearch** and designed a Maker-Checker document review flow with a **git-inspired version control system** for audit traceability.
- Created periodic web scrapers for financial data (stock news, reports) in **Node.js/Puppeteer and Python (BeautifulSoup, Selenium)**, persisting to **MongoDB**.
- Built a React frontend from scratch, giving Spark Capital search capabilities over financial documents, integrating audio streaming for conference call recordings via **AWS S3**.
- Shipped **6+ backend microservices** (Flask, AWS Cognito, S3) and **4+ React.js frontend MVPs**, including a serverless AWS Chalice app, owning end-to-end REST API design and frontend-backend integration.
- Mentored a team of 4 interns, conducting code reviews and providing structured guidance on engineering practices.

Engineering Intern

Sep 2018 - Mar 2019

Redcarpetup  (YC funded, credit & lending startup)

New Delhi, India (On-site)

- Delivered REST APIs and frontends for user onboarding flows, card delivery tracking, and admin portal features using **Flask, Postgres and React**, integrating BookMyPacket for delivery tracking.

PROJECTS

LiveDraft (Real-Time Collaborative Markdown Wiki)  github.com/princemathur4/LiveDraft

- Designed a real-time collaborative wiki using FastAPI, PostgreSQL, React, and Docker with **CRDT-based multi-user editing** over WebSockets (y-js), implementing JWT authentication, hierarchical page-tree, PostgreSQL ts-vector full-text search, and **XSS-safe markdown rendering**, document persistence and debounced autosave for reliability.

Question Answering Bot (RAG-based PDF Q&A System)  github.com/princemathur4/question-answering-bot

- Implemented a **Flask-based RAG pipeline** over PDFs using **OpenAI, LangChain, and Pinecone DB** for vector storage, enabling **document-grounded answers** with semantic retrieval.

EDUCATION

Woolf University (Scaler Neovarsity)

2026 (Expected)

M.S. in Computer Science

Manipal University

2018

BE/B.Tech in Electronics and Communication